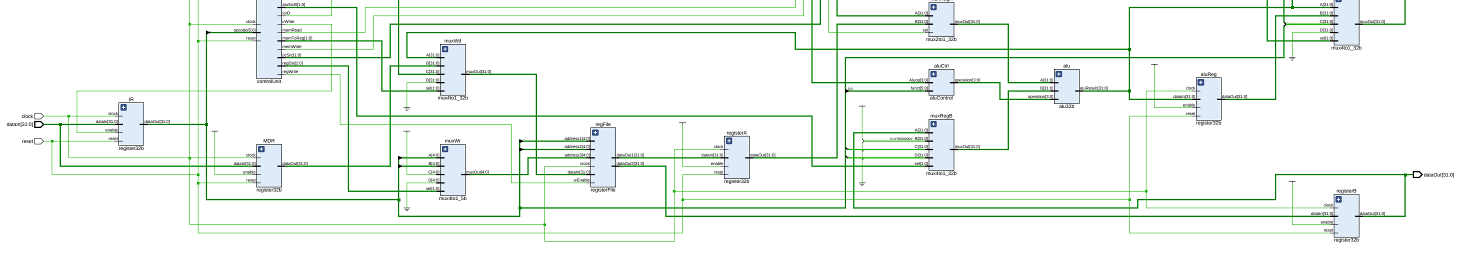
clock
dataIn[31:0]
reset
IR
clock
dataIn[31:0]
enable
reset
dataOut[31:0]
register32b
pc
clock
dataIn[31:0]
reset
dataOut[31:0]
register32b
controlU
aluOp[3:0]
aluSrcA
aluSrcB[1:0]
iorD
irWrite
memRead
memToReg[1:0]
memWrite
pcSrc[1:0]
regDst[1:0]
regWrite
clock
opcode[5:0]
reset
controlUnit
MDR
clock
dataIn[31:0]
enable
reset
dataOut[31:0]
register32b
muxWd
A[31:0]
B[31:0]
C[31:0]
D[31:0]
sel[1:0]
muxOut[31:0]
mux4to1_32b
muxWr
A[4:0]
B[4:0]
C[4:0]
D[4:0]
sel[1:0]
muxOut[4:0]
mux4to1_5b
regFile
address1[4:0]
address2[4:0]
address3[4:0]
clock
dataIn[31:0]
wEnable
dataOut1[31:0]
dataOut2[31:0]
registerFile
registerA
clock
dataIn[31:0]
enable
reset
dataOut[31:0]
register32b
muxRegA
A[31:0]
B[31:0]
sel
muxOut[31:0]
mux2to1_32b
aluCtrl
Aluop[3:0]
funct[5:0]
operation[3:0]
aluControl
5:0
muxRegB
A[31:0]
V=X"00000001" B[31:0]
C[31:0]
D[31:0]
sel[1:0]
muxOut[31:0]
mux4to1_32b
alu
A[31:0]
B[31:0]
operation[3:0]
aluResult[31:0]
alu32b
aluReg
clock
dataIn[31:0]
enable
reset
dataOut[31:0]
register32b
muxPc
A[31:0]
B[31:0]
sel
muxOut[31:0]
mux2to1_32b
muxPcSrc
A[31:0]
B[31:0]
C[31:0]
D[31:0]
sel[1:0]
muxOut[31:0]
mux4to1_32b
registerB
clock
dataIn[31:0]
enable
reset
dataOut[31:0]
register32b
address[31:0]
read
write
dataOut[31:0]